Mathematical Foundations of Political Methodology

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Day 3: Random Variables: Discrete

- Random Variables
- Expectation, Variance
- Discrete Probability Mass Function
- Cumulative Distribution Functions
- Joint Probability Mass Functions

Random Variables

Given a sample space Ω, and a probability law P:

Definition

A random variable is a function that assigns real numbers (usually) to events in a sample space Ω.

$$X(\omega) : \Omega \rightarrow \mathbb{R}$$

Random Variables: Example

Sample space from tossing three coins:

$$\Omega = \{HHH,\ HHT,\ HTH,\ THH,\ HTT,\ THT,\ TTH,\ TTT\}$$

Let $Y(\omega)$ be the number of heads in outcome $\omega \in \Omega$.

- $Y(HHH) = 3$
- $Y(HHT) = Y(HTH) = Y(THH) = 2$
- $Y(HTT) = Y(THT) = Y(TTH) = 1$
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Random Variables: Example II

Sample space from the number of dots on a die roll.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Let $Z(\omega)$ be the dots on the die face $\omega \in \Omega$.

- $Z(1) = 1$
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Random Variables: Example III

Sample space from the number of dots on a die roll.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Let $X(\omega)$ be 3 to the power of the number of dots on the die face $\omega \in \Omega$ minus 1.

- $X(1) = 3^{1-1} = 3^0 = 1$
- $X(2) = 3^{2-1} = 3$
- $X(3) = 3^{3-1} = 9$
- $X(4) = 3^{4-1} = 27$
- $X(5) = 3^{5-1} = 81$
- $X(6) = 3^{6-1} = 243$

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Let $X(\omega)$ be the number of dots to the 0 power -1.

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functions

- Random Variables are functions.
- In R, we see functions look like a word followed by ():
  - mean(), median(), head(), str()
- Our own functions are written using:

```
function(input){
    do something
    return(output)}
```

Properties of Random Variables

Random variables (for us) will always give a number.

- X,Y, Z: capital letters for a random variable.
- x, y, z: are the output of the function.

Kinds of Random Variables

Definition

A random variable may be

1. *discrete: if outcomes are countable.*
2. *continuous: If outcomes can be any value in some interval.*
3. *mixed: if outcomes are a combination of events and intervals.*

Connecting random variables to probability

- X assigns some numbers to events.
- Remember, probability assigns a likelihood to an event.
- What is the probability that $X(\omega)$ gives some number x?
- $P(X = x)$
- This is another function (of a function)

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- X assigns some numbers to events.
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- $P(X = x) \equiv P(\{\omega \in \Omega : X(\omega) = x\})$
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Discrete Probability Function

Definition

If X is a discrete random variable, the probability of each outcome x is provided by a probability function:

$$p_X(x) = P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$$

- this is also called the *probability mass function* (pmf)
- We often just write $p(x)$, some people use $f_X(x)$

Building a Discrete Probability Function: Example

- Sample space: $\Omega = \{\text{Pashto, Not Pashto}\}$
- X=1 if ω = Pashto
- X=0 if ω = Not Pashto
- P(X=1)=.3

A pmf can take on any value of x, so we must consider all possibilities of x.

- $x = 1$
- $x = 0$
- all other values of x.

- If $x = 1$

$$P(X = x) = P(X = 1) = P(\text{Pashto}) = p = .3$$

- If $x = 0$

$$P(X = x) = P(X = 0) = P(\text{Not Pashto}) = 1 - p = .7$$

- all other values of x.

$$P(X = x) = 0$$

Combining these results, we get the probability function.

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

This can be written shorthand as:

$$f_X(x) = p^x(1 - p)^{(1-x)}, \quad x \in \{0, 1\}$$

Where the 0 is left implicit.

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Dichotomous process

To summarize

- If X only has two values 0 and 1 and $p = P(X = 1)$ then X has the probability function:

$$f_X(x) = p^x(1-p)^{(1-x)}, \quad x \in \{0, 1\}$$

- This is called a Bernoulli distribution:

$$X \sim \text{Bernoulli}(p)$$

- p is a *parameter*

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Bernoulli Random Variable

Suppose we flip a fair coin and $Y = 1$ if the outcome is Heads .

$$\begin{aligned} Y &\sim \text{Bernoulli}(1/2) \\ p(1) &= (1/2)^1(1-1/2)^{1-1} = 1/2 \\ p(0) &= (1/2)^0(1-1/2)^{1-0} = (1-1/2) \end{aligned}$$

Example: Winning a War

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&= B \times p(Y = 1) - c(P(Y = 1) + P(Y = 0)) \\
&= B \times \pi - c
\end{aligned}$$

Example: Sequence of Bernoulli trials

We might be interested in the outcome of several independent trials **ie. with replacement**.

Consider a coin, flipped twice. Suppose $X_1(T) = 0$, $X_1(H) = 1$, $X_2(T) = 0$, $X_2(H) = 1$

$$P(X_1 = 1,\ X_2 = 1) = P(X_1 = 1 \cap X_2 = 1) = P(X_1 = 1)P(X_2 = 1)$$

$$p(x_1, x_2) = p^{x_1}(1-p)^{1-x_1}p^{x_2}(1-p)^{1-x_2}$$

$$p(x_1, x_2) = p^{x_1+x_2}(1-p)^{2-x_1-x_2}$$

Let $X_1, ..., X_n$ be independent Bernoulli random variables with the same parameter p.

$$p(x_1, ...x_n) = p(x_1)p(x_2)...p(x_n) = p^{x_1+...x_n}(1-p)^{n-x_1-....-x_n}$$

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Let $X_1, ..., X_n$ be independent Bernoulli random variables with the same parameter p.

$$p(x_1, ...x_n) = p(x_1)p(x_2)...p(x_n) = p^{x_1+...x_n}(1-p)^{n-x_1-....-x_n}$$

for $x_i \in \{0, 1\}$ and $i = 1, ..., n$

Example: Sequence of Bernoulli trials

We might be interested in the outcome of several independent trials **ie. with replacement**.

Consider a coin, flipped twice. Suppose $X_1(T) = 0$, $X_1(H) = 1$, $X_2(T) = 0$, $X_2(H) = 1$

$$P(X_1 = 1,\ X_2 = 1) = P(X_1 = 1 \cap X_2 = 1) = P(X_1 = 1)P(X_2 = 1)$$

$$p(x_1, x_2) = p^{x_1}(1-p)^{1-x_1}p^{x_2}(1-p)^{1-x_2}$$

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In words: for all values of x with $p(x)$ greater than zero, take the weighted average of the values

Expectation informally

The expectation is based on the principle that the value of a future gain should be directly proportional to the chance of getting it.

$$E[X+Y]=E[X]+E[Y]$$

$$E[a]=a$$

$$E[aX]=aE[X]$$

$$E[E[X]]=E[X]$$

$$E[XY]\neq E[X]\times E[Y]$$

Properties of the Expecation

$$E[a] = a$$

Proof: Suppose Y is a random variable = a with probability 1 and 0 otherwise:

$$E[Y] = \sum_{y:p(y)>0} yp(y) = a * 1$$

Also:

$$E[aX] = \sum_{x:p(x)>0} a \times p(x)$$

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Proof.

$$E[I] = 1 \times P(A) + 0 \times P(A^c)$$

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Expected value is a measure of central tendency.

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&= E[X^2] - E[X]^2 \\
&= \text{Var}(X)
\end{aligned}$$

Variance

Definition

The variance of a random variable X, $var(X)$, is

$$
\begin{aligned}
var(X) &= E[(X - E[X])^2] \\
&= E[X^2] - E[X]^2
\end{aligned}
$$

- We will define the standard deviation of X, $\text{sd}(X) = \sqrt{\text{var}(X)}$
- $\text{var}(X) \geq 0$.

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Corollary

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&= a^2(E[X^2] - E[X]^2) \\
&= a^2Var(X)
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Bernoulli Random Variable Moments

Suppose $Y \sim \text{Bernoulli}(\pi)$

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$E[Y] = \pi$
$\text{var}(Y) = \pi(1 - \pi)$ What is the maximum variance?

Trials with More than Two Outcomes

Definition

Suppose we observe a trial, which might result in J outcomes.
And that $P(outcome\ = i) = \pi_i$
$\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_J)$ *where* $Y_j = 1$ *if outcome j occurred and 0 otherwise.*
Then $\boldsymbol{Y}$ *follows a multinomial distribution, with*

$$p(\boldsymbol{y}) \quad = \quad \pi_1^{y_1}\pi_2^{y_2}\ldots\pi_k^{y_k}$$

if $\sum_{i=1}^{k} y_i = 1$ *and the pmf is* 0 *otherwise.*
Equivalently, we'll write

$$\begin{aligned} \boldsymbol{Y} &\sim \mathit{Multinomial}(1, \boldsymbol{\pi}) \\ \boldsymbol{Y} &\sim \mathit{Categorial}(\boldsymbol{\pi}) \end{aligned}$$

Multinomial Properties + Notes

Computer scientists: commonly call Multinomial$(1, \boldsymbol{\pi})$ Discrete$(\boldsymbol{\pi})$.

$$\begin{aligned} E[X_i] &= N\pi_i \\ \text{var}(X_i) &= N\pi_i(1 - \pi_i) \end{aligned}$$

Counting the Number of Events

Often interested in counting number of events that occur:

1) Number of wars started
2) Number of speeches made
3) Number of bribes offered
4) Number of people waiting for license

Generally referred to as event counts

Stochastic processes: a course provide introduction to many processes (Queing Theory)

Definition

Suppose X is a random variable that counts the number of successes in N independent and identically distributed Bernoulli trials. Then X is a *Binomial* *random variable,*

$$p(k) \;=\; \binom{N}{k}\pi^k(1-\pi)^{1-k}$$

for $k \in \{0, 1, 2, \ldots, N\}$ and $p(k) = 0$ otherwise.
Equivalently,

$$Y \;\sim\; Binomial(N, \pi)$$

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$E[Z] = N\pi$
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Example: Counts

We might be interested in the *sum* of several independent trials:

- n draws from a population size N with replacement.
- If Y is number of the sample that have a binary characteristic (support Trump, show up heads, etc).
- and M is the number of the people in the population that have a binary characteristic.

Our random variable, Y, is equal to the sum of successes. Let y be an integer between 0 and n,

$$P(Y = y) = \binom{n}{y} p^y (1-p)^{(n-y)}$$

where $p = \frac{M}{N}$

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- A model to count the number of successes across N trials
  - Assume the Bernoulli trials are independent
  - Each Bernoulli trial i is

$$Y_i \sim \text{Bernoulli}(\pi)$$

    Independent and identically distributed.
- Z = number of successful trials
- Derive probability mass function $P(Z = M) = p(M)$
- One way to obtain M successful trials:

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Origins of Binomial

One head on one Coin:

$$P(Y = 1) = p$$

One head on two coins:

$$P(Y = 1) = p(1 - p) + (1 - p)p = 2 * p(1 - p)^1$$

One head on three coins:

$$P(Y = 1) = p(1-p)(1-p)+(1-p)p(1-p)+(1-p)(1-p)p = 3*p(1-p)^2$$

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Three heads on four coins

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Origins of Binomial

If $X_1, ... X_n$ are Bernoulli distributed with probability p, then

$$Y = \sum_{k=1}^{n} X_k \sim \text{Binomial}(n, p)$$

n and p are parameters of the binomial.

Assumptions of Binomial Distribution

- The number of trials is fixed,
- The probability of success is the same for each trial.
- The trials are independent

Example: Baseball!

- The World Series of Baseball ends when one team wins 4 games.
- Each game is a Bernoulli trail.
- Length of World Series $\in \{4,\ 5,\ 6,\ 7\}$.
- What is the probability that the World Series lasts 7 games?
- P(X=7)?

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Example: Counts in a time period

We might be interested in the sum of events occurring in a specific time or space.

- Number of wars in a year.
- Number of diseased villages in a district.

Let X be the number of times an event occurs in a unit of time or space.

- The probability that the event occurs in a given unit of time or space is the same for all the units.
- The number of events that occur in one unit of time is independent of the number of events in other units.

Then X is a Poisson random variable, with rate $= \lambda$ and pf:

$$p(x) = \frac{\lambda^x}{x!} e^{-\lambda} \quad x = 0, 1, 2...$$

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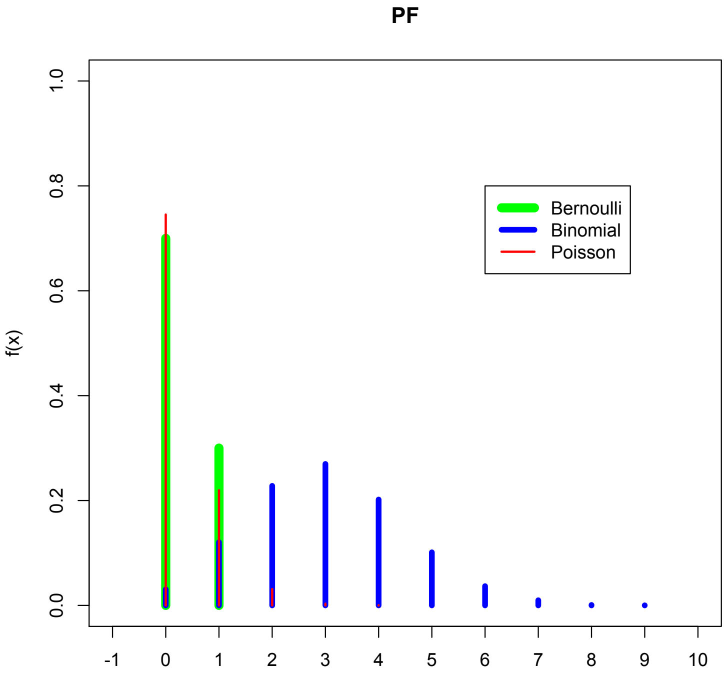

PF
f(x)
0.0
0.2
0.4
0.6
0.8
1.0
-1
0
1
2
3
4
5
6
7
8
9
10
Bernoulli
Binomial
Poisson

Functions of Random Variables

Definition

If $Y = g(X)$, and X is a random variable with probability mass function $p_X(x)$, Y is a random variable with pmf:

$$p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$$

Cumulative Distribution Function

Definition

The Cumulative Distribution Function (CDF) of a random variable X is:

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty$$

The CDF characterizes how probability cumulates as X gets larger.

Cumulative Distribution Function and the Probability Function

Definition

The Cumulative Distribution Function (CDF) of a random variable X is:

$$F_X(x) = \sum_{k \leq x} p_X(k)$$

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10,\ p = .3)$ What is the probability that $x \leq 4$?

$$F_X(4) = P(X \leq 4)$$

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Cumulative Distribution Function

Theorem

F(x) is a CDF if and only if:

1. $F_X(x)$ *is nondecreasing as x increases:*

$$x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$$

2. $F_X(x)$ *has limits at* $\pm\infty$

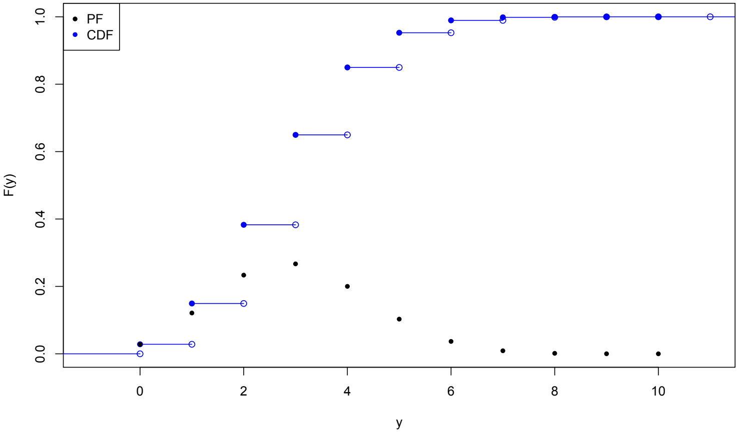
$$\lim_{x \to -\infty} F_X(x) = 0$$

$$\lim_{x \to \infty} F_X(x) = 1$$

3. *Continuity from the right:*

$$F_X(x) = \lim_{y \to x,\ y > x} F(x) \quad \forall x$$

CDF



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We might (or often) apply a function to a random variable $g(X)$.

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Expected value of a function of a random variable: Suppose X is a discrete random variable that takes on values x_i, $i = \{1, 2, \ldots,\}$, with probabilities $p(x_i)$.

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$$E[g(X)] = \sum_i g(x_i)p(x_i)$$

Functions of Random Variables: Corollary

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Suppose X is a random variable and a and b are constants (not random variables). Then,

$$E[aX + b] = aE[X] + b$$

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Inequalities

Definition

Markov's Inequality Let X be a random variable, and g be a nonnegative function, then for any positive a:

$$P(g(X) \geq a) \leq \frac{E[g(X)]}{a}$$

Definition

Jensen's Inequality Let X be a random variable with $E(|X|) < \infty$. If g is convex, then

$$E[g(X)] \geq g(E(X))$$

Example of Chebychev

If X is a random variable, with mean μ and variance σ^2:

$$P(|X - \mu| \geq 2\sigma) \leq .25$$

$$P(|X - \mu| \geq 2\sigma) = P(|X - \mu|^2 \geq 4\sigma^2)$$

$$g(X) = |X - \mu|^2$$

$$E(g(X)) = \sum |X - \mu|^2 p_x = Var(X) = \sigma^2$$

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Famous Distributions

- Bernoulli
- Binomial
- Multinomial
- Poisson

Models of how world works.

Poisson Distribution

Definition

Suppose X *is a random variable that takes on values* $X \in \{0, 1, 2, \ldots, \}$ *and that* $P(X = k) = p(k)$ *is,*

$$p(k) \quad = \quad e^{-\lambda} \frac{\lambda^k}{k!}$$

for $k \in \{0, 1, \ldots, \}$ *and* 0 *otherwise. Then we will say that* X *follows a Poisson distribution with rate parameter* λ*.*

$$X \quad \sim \quad Poisson(\lambda)$$

Poisson Distribution

Properties:

1) It is a probability distribution.

Poisson Distribution

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Very useful distribution.

Joint PMFs

If we have more than one random variable, we can study their joint behavior.
If (x, y) are possible values of random variables X and Y, then

$$p_{X,Y}(x, y) = P(X = x,\ Y = y)$$

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Joint PMFs

If we want to know the probability of a set of joint events $(x, y) \in A$

$$P((X, Y) \in A) = \sum_{(x,\ y) \in A} p_{X,Y}(x,\ y)$$

We can also calculate the pmfs of X and Y individually (these are the marginal distributions:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Example of Joint PMF

$$\Omega_1 = \{\text{Non-Democracy}, \quad \text{Democracy}\}$$

$$X(\omega) = \begin{cases} 0 \text{ if } \omega = \text{Non-Democracy} \\ 1 \text{ if } \omega = \text{Democracy} \end{cases}$$

$$\Omega_2 = \{\text{not-Killed}, \quad \text{Killed}\}$$

$$Y(\omega) = \begin{cases} y = 0 \text{ if } \omega = \text{not Killed} \\ y = 1 \text{ if } \omega = \text{Killed} \end{cases}$$

$p_{X,Y}$	0	1	p_Y
1	0.034	0.005	0.039
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$$E[X+Y] = E[X] + E[Y]$$

Proof:

$$E[X+Y] = \sum_x \sum_y (x+y)p(x,y)$$

$$= \sum_x \sum_y xp(x,y) + \sum_x \sum_y yp(x,y)$$

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Conditional Probability Distribution Function

Definition

Suppose X and Y are discrete random variables with joint pdf $p(x, y)$. Then define the conditional probability function $p(x|y)$ as

$$p(x|y) = \frac{p(x, y)}{p_Y(y)}$$

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Begin with discrete case.

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$$p(x_j) = \sum_{i=1}^{N} p(x_j|y_i)p(y_i)$$

A Table

	$Y = 0$	$Y = 1$	
$X = 0$	p(0,0)	p(0, 1)	$p_X(0)$
$X = 1$	p(1,0)	p(1,1)	$p_X(1)$
	p_Y (0)	p_Y (1)	

A Table

	Y = 0	Y = 1	
X = 0	0.01	0.05	?
X = 1	0.25	0.69	?
	0.26	0.74	

$$p_X(0) = p(0|y=0)p(y=0) + p(0|y=1)p(y=1)$$

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$$p_X(1) = p(1|y=0)p(y=0) + p(1|y=1)p(y=1)$$

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If X and Y are not independent, we will say they are dependent

Conditional Distribution

If X and Y are independent, then

$$\begin{aligned} p_{X|Y}(x|y) &= \frac{p(x,y)}{p_Y(y)} \\ &= \frac{p_X(x)p_Y(y)}{p_Y(y)} \\ &= p_X(x) \end{aligned}$$

In words: the distribution of X does not change as levels of Y change.

Conditional Expectation

$$\mathbb{E}(Y|X) = \sum_{y} y \cdot p_{X|Y}(x|y)$$

$\mathbb{E}(Y|X)$ is the *best* way to predict Y given X.